

[Molybdenum: Fact Sheet for Consumers \(original version\)](#)

# Molybdenum: Fact Sheet for Consumers

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## What is molybdenum and what does it do?

Molybdenum is a mineral that you need to stay healthy. Your body uses molybdenum to process proteins and genetic material like DNA. Molybdenum also helps break down drugs and toxic substances that enter the body.

## How much molybdenum do I need?

The amount of molybdenum you need depends on your age. Average daily recommended amounts are listed below in micrograms (mcg).

| Life Stage                    | Recommended Amount |
|-------------------------------|--------------------|
| Birth to 6 months             | 2 mcg              |
| Infants 7–12 months           | 3 mcg              |
| Children 1–3 years            | 17 mcg             |
| Children 4–8 years            | 22 mcg             |
| Children 9–13 years           | 34 mcg             |
| Teens 14–18 years             | 43 mcg             |
| Adults 19 years and older     | 45 mcg             |
| Pregnant teens and women      | 50 mcg             |
| Breastfeeding teens and women | 50 mcg             |

## What foods provide molybdenum?

Many foods contain molybdenum. The amount of molybdenum in food depends on the amount of molybdenum in the soil and in the water used for irrigation. You can get recommended amounts of molybdenum by eating a variety of foods, including the following:

- Legumes such as black-eyed peas and lima beans
- Whole grains, rice, nuts, potatoes, bananas, and leafy vegetables

- Dairy products, like milk, yogurt, and cheese
- Beef, chicken, and eggs

## What kinds of molybdenum dietary supplements are available?

Some multivitamin/mineral supplements contain molybdenum. Other supplements contain molybdenum alone or together with other minerals.

## Am I getting enough molybdenum?

Most people in the United States get enough molybdenum from the foods they eat.

## What happens if I don't get enough molybdenum?

Molybdenum deficiency is very rare in the United States. It happens only in people with a very rare genetic disorder called molybdenum cofactor deficiency. This disorder prevents the body from using molybdenum. It can cause seizures and severe brain damage that usually leads to death within days after birth.

## What are some effects of molybdenum on health?

Whether molybdenum affects any disease or health condition isn't known.

## Can molybdenum be harmful?

Molybdenum from food and beverages doesn't cause any harm. However, people exposed to high levels of molybdenum in the air and soil, such as miners and metalworkers, sometimes develop achy joints, gout-like symptoms, and high blood levels of uric acid (a substance that is normally excreted in your urine).

The daily upper limits for molybdenum include intakes from all sources—food, beverages, and supplements—and are listed below in micrograms (mcg).

| Ages                | Upper Limit     |
|---------------------|-----------------|
| Birth to 12 months  | Not established |
| Children 1–3 years  | 300 mcg         |
| Children 4–8 years  | 600 mcg         |
| Children 9–13 years | 1,100 mcg       |
| Teens 14–18 years   | 1,700 mcg       |
| Adults              | 2,000 mcg       |

## Does molybdenum interact with medications or other dietary supplements?

Molybdenum is not known to interact or interfere with any medicines, but it's always important to tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if these dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients, such as molybdenum.

## Molybdenum and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

## Where can I find out more about molybdenum?

- For more information on molybdenum
  - Office of Dietary Supplements (ODS) [Health Professional Fact Sheet on Molybdenum](#)
- For more advice on choosing dietary supplements
  - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
  - [Dietary Guidelines for Americans](#)

## Glossary

### absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

### deficiency

An amount that is not enough; a shortage.

### deoxyribonucleic acid

DNA. The molecules inside cells that carry genetic information and pass it from one generation to the next.

### dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

### Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

### dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

### disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

### drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

### fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

### gene

The functional and physical unit of heredity passed from parent to offspring. Genes are pieces of DNA, and most genes contain the information for making a specific protein.

### genetic disorder

A disease or disorder caused by an alteration or variation (mutation) in a gene or group of genes in the cells of an individual. Examples of genetic disorders include breast cancer, cystic fibrosis, Parkinson's disease, and celiac disease. They can be inherited or can occur without a known cause.

infant

A child younger than 12 months old.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

legume

Dried beans and peas, including kidney beans, pinto beans, black beans, navy beans, lima beans, black-eyed peas, garbanzo beans (chickpeas), split peas, and lentils. Legumes are good sources of protein, iron, zinc, dietary fiber, folate, and potassium.

microgram

$\mu\text{g}$  or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

seizure

Sudden changes in behavior caused by excessive electrical activity in the brain.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

toxic

Having to do with poison or something harmful to the body. Toxic substances usually cause unwanted health effects.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

urine

Excess liquids and wastes that have been filtered from the blood by the kidneys, stored in the bladder, and removed from the body through the urethra (the tube that carries urine from the bladder to outside the body).

**vitamin**

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

**whole grain**

Unprocessed seeds of edible grasses, including brown rice, buckwheat, bulgur, millet, popcorn, oats, quinoa, whole-grain barley, whole rye, whole wheat, and wild rice. Grains that are ground, cracked, or flaked can be labeled whole grain if they have the same amount of bran, germ, and endosperm (the inner part of the seed kernel) as the intact grain. Whole grains are sources of iron, magnesium, selenium, B vitamins, and dietary fiber. Eating whole grains may help lower the risk of heart disease, obesity, and type 2 diabetes.

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